

Instance — Nested UQ hyperparameter of (2) with explicit Ass. concentration

research/formal (working)

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Selected object = $\hat{\theta}/\tilde{\theta}$ from (2) only. Specialization of writeup (2) for UQ hyperparameter selection (not CREDO decision z). Closes the open gap in `part_i_randomized_design.tex`: explicit $\varepsilon(n), \nu(n)$ for Ass. concentration. CRC nested structure is used only to warrant fixed- θ validity (FV)/(1); it is not imported as a CREME algorithm.

1 Instance of (2)

Let $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$ be i.i.d. from P , and let $s(x, y) \in [0, 1]$ be a fixed score (trained on a disjoint proper-training split if desired; treated as nonrandom w.r.t. $\mathcal{D}$ below). For $\theta \in \Theta := [0, 1]$ define the nested sets

$$\mathcal{C}_\theta(x) := \{y : s(x, y) \leq \theta\}. \quad (1)$$

Monotonicity: $\theta \leq \theta' \Rightarrow \mathcal{C}_\theta(x) \subseteq \mathcal{C}_{\theta'}(x)$.

Definition 1 (Empirical miscoverage objective). Write $s_i := s(X_i, Y_i)$ and

$$\hat{f}(\theta) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Y_i \notin \mathcal{C}_\theta(X_i)\} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{s_i > \theta\} = 1 - \hat{F}_n(\theta),$$

where $\hat{F}_n$ is the empirical CDF of $\{s_i\}$. Population counterpart: $f^*(\theta) := \mathbb{P}\{s_1 > \theta\} = 1 - F(\theta)$.

Definition 2 (Box constraints = trivial $\hat{g}$). Take $\hat{g}(\theta) := (-\theta, \theta-1)$ so $\Theta_0 = \{\theta : \hat{g}(\theta) \leq 0\} = [0, 1]$ pathwise. Oracle match: $g^* \equiv \hat{g}$, hence $\Theta_0 = \{g^* \leq 0\}$ holds with probability 1 (no feasible-set estimation error).

The deterministic program is then

$$\hat{\theta}(\mathcal{D}) \in \arg \min_{\theta \in [0, 1]} \hat{f}(\theta). \quad (2_{\text{nest}})$$

Note: $\hat{f}$ is nonincreasing, so a measurable tie-break yields $\hat{\theta} = 1$ a.s. Randomized designs of `part_i_randomized_des` (RNM on a grid / soft-argmin) still act on the score vector $\hat{f}|_{\Theta_m}$ and are the objects of interest for goal (iii).

Remark 1 ((FV)/(1) warrant — NOT for all $\theta \in [0, 1]$). Writeup (1) is an assumption for each prespecified θ in the domain where the family is valid. For the score-threshold sets (1),

$$\mathbb{P}\{Y \notin \mathcal{C}_\theta(X)\} = 1 - F(\theta),$$

so (1) at level α holds iff $\theta \geq F^{-1}(1 - \alpha)$ (population), not for arbitrary $\theta \in [0, 1]$. **Do not** cite CRC as giving (1) for every fixed threshold; CRC controls a data-driven threshold, which is not our selection map. Operational domain for Part II: keep randomized $\tilde{\theta}$ (and oracle A_0) supported on a valid set, e.g. $\Theta_{\text{val}} = [t_\alpha, 1]$ with $t_\alpha = F^{-1}(1 - \alpha)$, or invoke oracle-validity (OV) only for that A_0 (Part II needs (OV), not a blanket $\sup_{\theta \in [0, 1]}$ form of (FV)).

2 Ass. concentration with explicit rates

Lemma 1 (DKW concentration for nested miscoverage). Fix $\nu \in (0, 1)$. Under i.i.d. scores $s_i \in [0, 1]$,

$$\varepsilon(n, \nu) := \sqrt{\frac{\log(2/\nu)}{2n}}, \quad \nu(n) := \nu \quad (2)$$

satisfy Ass. concentration of `part_i_randomized_design.tex` (exact feasible-match special case: box $\hat{g}$):

$$\mathbb{P}\left\{\|\hat{f} - f^*\|_{\infty, [0, 1]} \leq \varepsilon(n, \nu), \Theta_0 = \{g^* \leq 0\}\right\} \geq 1 - \nu.$$

Proof: $\Theta_0 = \{g^* \leq 0\}$ always (Def. 2). $\|\hat{f} - f^*\|_\infty = \|\hat{F}_n - F\|_\infty$, and the Dvoretzky–Kiefer–Wolfowitz inequality gives $\mathbb{P}\{\|\hat{F}_n - F\|_\infty > \varepsilon\} \leq 2e^{-2n\varepsilon^2}$; set the right-hand side to ν and solve for ε .

Corollary 1 (Finite grid — Hoeffding + union bound). If instead $\Theta_m = \{\theta_1, \dots, \theta_m\} \subset [0, 1]$ and (2) is restricted to Θ_m , then

$$\varepsilon_{\text{grid}}(n, \nu; m) := \sqrt{\frac{\log(2m/\nu)}{2n}} \quad (3)$$

satisfies $\|\hat{f} - f^*\|_{\infty, \Theta_m} \leq \varepsilon_{\text{grid}}$ with probability at least $1 - \nu$ (Hoeffding on each coordinate + union bound). Same trivial $\hat{g} / \Theta_0 = \Theta_m$.

3 Plug-in to randomized design (goal (iii))

Corollary 2 (RNM / soft rates with concrete ε, ν). On the good event of Lemma 1 (or Cor. 1), take $b = 2\varepsilon(n, \nu)/\eta$ (report-noisy-min) or $\beta = \eta/(2\varepsilon(n, \nu))$ (soft-argmin) on a finite net of the **valid** domain $\Theta_{\text{val}} = [t_\alpha, 1]$ with $t_\alpha = F^{-1}(1 - \alpha)$ (Remark 1) — not on all of $[0, 1]$. Then Lemmas rnm / soft of `part_i_randomized_design.tex` yield $(\eta, 0, \nu)$ -stability of $\tilde{\theta}$, with $\eta > 0$ a free design knob. Part II Infl at fixed classical level then applies because oracle soft / RNM on Θ_{val} satisfies (OV) at $q = \alpha$: $\text{Infl} \leq (e^\eta - 1)\alpha + \nu$, which $\rightarrow \nu$ (then $\nu \downarrow 0$ with n) as $\eta \downarrow 0$. Anti-claim: the same design on a net of full $[0, 1]$ is still $(\eta, 0, \nu)$ -stable, but (OV)/(FV) fail as $\eta \rightarrow 0$ (mass escapes below t_α), so the Infl bound is **not** claimed there.

Remark 2 (What is not claimed). No claim that deterministic $(\mathcal{Q}_{\text{nest}})$ is an interesting radius selector (it collapses to $\theta = 1$). No CREDO z , no CRC $\hat{\lambda}$ as live $\hat{\theta}$, no recalibration of $\mathcal{C}$.